Econ 326
Assignment 6

1. An econometrician used R to regress the dependent variable `liver` on the independent variable `alcohol`, where `liver` is the number of liver disease deaths per 100,000 people in a country, and `alcohol` is consumption of alcohol in liters per capita in a country. The following output was obtained:

```
> model <- lm(liver ~ alcohol, data = mydata)
> summary(model)
...
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  10.8548     2.8024   3.874 0.001030 **
alcohol       3.5864     0.7541       A        B
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.29 on 19 degrees of freedom


> confint(model)
               2.5 %   97.5 %
(Intercept) 4.989313 16.72033
alcohol            C        D
```

   (a) Several entries in the output were replaced with letters. Find A–D. Show your work.

   (b) Test at 5% significance level that the coefficient of `alcohol` is 5 (against the alternative that it is different from 5).

   (c) Test the same hypothesis as in part (b) at 10% significance level.

2. An econometrician used R to regress the dependent variable `ln_heart` on the independent variable `ln_alcohol`, where `ln_heart` is the natural logarithm of the number of heart disease deaths per 100,000 people in a country, and `ln_alcohol` is the natural logarithm of consumption of alcohol in liters per capita in a country. The following output was obtained:

```
> model <- lm(ln_heart ~ ln_alcohol, data = mydata)
> summary(model)
...
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  5.361366   0.132482  40.467   <2e-16 ***
ln_alcohol          A          B       C        D
...

> confint(model)
```

```
                  2.5 %     97.5 %
(Intercept)    5.084078   5.638654
ln_alcohol    -0.611437  -0.095217
```

(a) Several entries in the output were replaced with letters. Find A–D if the number of observations was 21. Provide careful explanations for your answers.

(b) What is the interpretation of the coefficient on `ln_alcohol`? Explain.

(c) Test at 5% significance level $H_0 : \beta \geq 0$ against $H_1 : \beta < 0$, where β is the coefficient on `ln_alcohol`.

3. An econometrician used R to regress `y` on `x`. The following output was obtained:

```
> model <- lm(y ~ x, data = mydata)
> summary(model)
...
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 3.109574   0.182132  17.073   <2e-16 ***
x           0.304925   0.177374       A        B
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.91 on 23 degrees of freedom
Multiple R-squared:  0.1139, Adjusted R-squared:  0.0753
F-statistic: 2.96 on 1 and 23 DF,  p-value: 0.09900

> confint(model)
                 2.5 %    97.5 %
(Intercept)   2.732806  3.486343
x                    C         D
```

(a) Find A–D. Provide careful explanations for your answers.

(b) Let β denote the coefficient on `x`. Test at 5% significance level $H_0 : \beta = 0$ against $H_1 : \beta \neq 0$.

(c) Test at 5% significance level $H_0 : \beta \leq 0$ against $H_1 : \beta > 0$.

(d) Explain the difference between the outcomes in parts (b) and (c).

4. Consider testing the null hypothesis $H_0 \colon \beta = 1$ at the 5% significance level using two different tests:

- **Two-sided test:** $H_0 \colon \beta = 1$ vs. $H_1 \colon \beta \neq 1$
- **One-sided test:** $H_0 \colon \beta = 1$ vs. $H_1 \colon \beta > 1$

Suppose that $\mathrm{Var}(\hat{\beta}) = 4$ (known), so $\sigma_{\hat{\beta}} = 2$. Since the variance is known, use the standard normal distribution.

(a) Find the rejection regions for both tests in terms of $\hat{\beta}$.

(b) Compute the power of the two-sided test for each of the following true values of β: $\beta^* \in \{-3,\ 0,\ 2,\ 5\}$.

(c) Compute the power of the one-sided test for each of the same true values: $\beta^* \in \{-3,\ 0,\ 2,\ 5\}$.

(d) Compare the power of the two tests. When does the one-sided test have higher power? When does it have lower power? Explain.

5. Consider a two-sided test of $H_0\colon \beta = 0$ against $H_1\colon \beta \neq 0$ at the 5% significance level. Assume the variance of $\hat{\beta}$ is known, so the standard normal distribution applies.

When $\beta^* > 0$, the power of the two-sided test is approximately

$$\text{Power} \approx 1 - \Phi\left(z_{0.975} - \frac{\beta^*}{\sigma_{\hat{\beta}}} \right),$$

where Φ is the standard normal CDF and $z_{0.975} = 1.96$. This approximation ignores the small probability of rejecting in the left tail, which is negligible when $\beta^* > 0$.

Suppose a researcher wants the test to have at least 80% power. The smallest value of $|\beta^*|$ for which this holds is called the *minimum detectable effect* (MDE): any true effect smaller than the MDE is unlikely to be detected by the test.

Hint: $\Phi^{-1}(0.20) = -0.84$, equivalently $z_{0.80} = 0.84$.

(a) Find the MDE when $\sigma_{\hat{\beta}} = 3$.

(b) Find the MDE when $\sigma_{\hat{\beta}} = 1.5$.

(c) Express the MDE as a general formula in terms of $\sigma_{\hat{\beta}}$. What does this relationship imply about how sample size affects the ability to detect small effects?

6. (a) Let X be a continuous random variable with a strictly increasing CDF F. Show that $U = F(X)$ has a Uniform$(0, 1)$ distribution.

Hints: (i) A random variable U has a Uniform$(0, 1)$ distribution if and only if $P(U \leq u) = u$ for all $u \in (0, 1)$. (ii) Because F is strictly increasing, its inverse F^{-1} exists and satisfies $F(x) \leq u \iff x \leq F^{-1}(u)$ and $F(F^{-1}(u)) = u$.

(b) Let $Z \sim N(0, 1)$ and define $V = 2\big(1 - \Phi(|Z|)\big)$, where Φ is the standard normal CDF. Show that $V \sim$ Uniform$(0, 1)$.

Hints: (i) Start by computing $P(V \leq v)$ for $v \in (0, 1)$. (ii) Note that $\Phi^{-1}(1 - v/2) = z_{1-v/2}$ is the critical value of a two-sided test at significance level v.

(c) Consider testing $H_0\colon \beta = \beta_0$ against $H_1\colon \beta \neq \beta_0$ with known variance $\sigma^2_{\hat{\beta}}$. Under H_0, the test statistic

$$Z = \frac{\hat{\beta} - \beta_0}{\sigma_{\hat{\beta}}}$$

has a standard normal distribution. The p-value of the two-sided test is

$$p = 2\big(1 - \Phi(|Z|)\big).$$

Using Part (b), show that the p-value has a Uniform$(0, 1)$ distribution under H_0.

(d) A test rejects H_0 whenever $p < \alpha$, where $\alpha \in (0, 1)$. Using the resul from Part (c), show that the size of this test (i.e., $P(\text{reject } H_0 \mid H_0 \text{ is true})$) equals α.